

Fig. 4: Gestures to control the movement of the robot

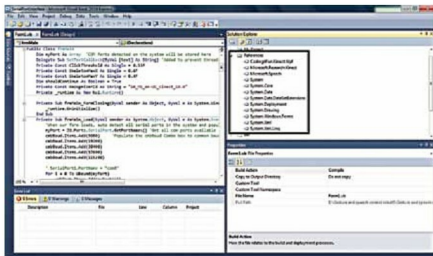


Fig. 5: Code compilation and debugging

depends on the dictionary used. Input is from a microphone array.

The main components are the RGB camera, depth sensor and microphone array. The depth sensor combines an IR laser projector with a monochrome CMOS sensor to get 3D video data. Besides these, there is a motor to tilt the sensor array up and down for the best view of the scene, and an accelerometer to sense position.

Robot. Fig. 3 shows the circuit of the robot. The robot is built around ATmega16 MCU (IC2), driver IC MAX232 (IC1), regulator IC 7805 (IC4), motor driver IC L293D (IC3) and a few discrete components.

COM port is connected to the computer using the USB-to-serial converter. Controlling commands to the robot are sent via serial port and the levels converted into 5V TTL/CMOS type by IC1. These TTL/CMOS signals are directly fed to the MCU (IC2) for controlling motors M1 and M2 to move

the robot in all directions. Port pins PB4 through PB7 of IC2 are connected to input pins IN1 through IN4 of IC3, respectively, to give driving inputs. EN1 and EN2 are connected to V_{CC} to keep IC3 always enabled. LED1 and LED2 are connected to ports PB1 and PB2 of IC2 for testing purpose.

Working of the project is simple. The robot is controlled via serial port and the controlling commands sent through the computer. These commands are generated by a software application running on the computer. This application interprets the gestures and sends corresponding commands to the robot through serial port. Each command initiates a process as shown in Table I.

If the operator stands in front of Kinect sensor at a minimum distance of 180 cm (or about 6 feet) and raises right hand up, the Visual Basic (VB) based application running in the computer interprets this gesture and

TABLE I
Gestures and Corresponding Commands

Gesture	Instruction	Process
Right hand up	s	Forward
Right hand swipe	r	Right
Left hand swipe	l	Left
Left hand up	w	Stop
Left hand back	v	Reverse
	a	LED1 'on'
	b	LED2 'on'

send 's' to the serial port. The robot is programmed to move forward if it receives 's' from the serial port. Similarly, for other gestures, corresponding letters as listed in the table are sent through the serial port to the robot.

Software

This gesture-controlled robot uses two software: a VB application running on the computer to interpret the gestures and a BASCOM program for the microcontroller to process the input signals and control the robot.

Visual Basic application. The software uses skeletal models and monitors joints to detect and interpret gestures. The analysis here is done using the position and orientation of joints and the relationship between each one of them (for example, the angle between joints and the relative position or orientation).

Advantages of using skeletal models are:

1. Algorithms are faster because only key parameters are analysed
2. Pattern matching against a template database is possible
3. Using key points allows the detection program to focus on significant parts of the body

The software solution is built using Microsoft Visual Studio 2010 Express. To build the program again and make your own executable file, download the source code and open the solution file from 'Gesture controlled robot\VB Application Program\SerialPortInterface' using Microsoft Visual Studio 2010 Express. Before building the solution, make sure that the references in